RAW MATERIALS / EQUIPMENT AND INSTALLATIONS.

SANDS :

MINING OF SANDS

MINING OF SANDS

Stock area of the supplier with sand to supply us.

Natural Sand: Fine natural sand of silicic origin.
Material free of clays and with a high sand equivalent. (0 to 2 mm)

MINING OF SANDS

Natural sand washed: Natural sand washed very fine of silicic origin. Material free of clays and with a high sand equivalent. (0 to 2 mm)

Sand washed from river : Sand from natural gravel pits. Lower content in fines and a high equivalent of sand. This type of aggregate concentrates a greater quantity of quartz and stone of great hardness. (0 to 3mm and 0 to 5mm)

MINING OF SANDS

CRUSHING SAND.

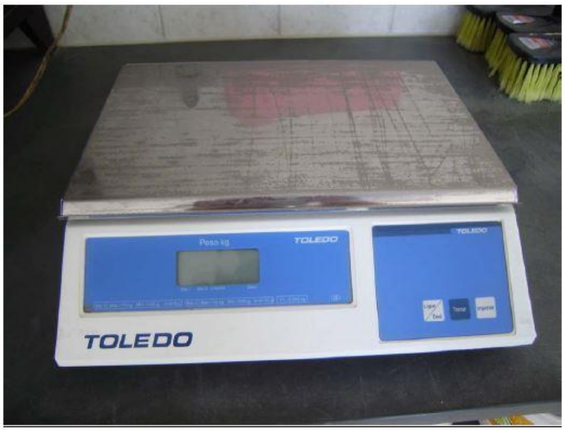
Pit sand: Sand from limestone rocks.
Material free of clays and with a high sand equivalent. (0 to 2 and 0 to 5 mm)

Gravel Eye Partridge: Sand from the crushing of limestone aggregates.
Exempt of clays. Live singing edges. Partridge eye from 3 to 5 mm.

<u>GRANULOMETRY</u>

Equipment to perform the "Granulometry”

<u>Precision scale.</u>



<u>Stove or fire cooking.</u>

<u>Sifting with different sieves.</u>



GRANULOMETRY.

The minerals in the sand no problem

Feldspar, Mica, Iron pyrite, Iron oxide, Pyroxenes / Amphiboles

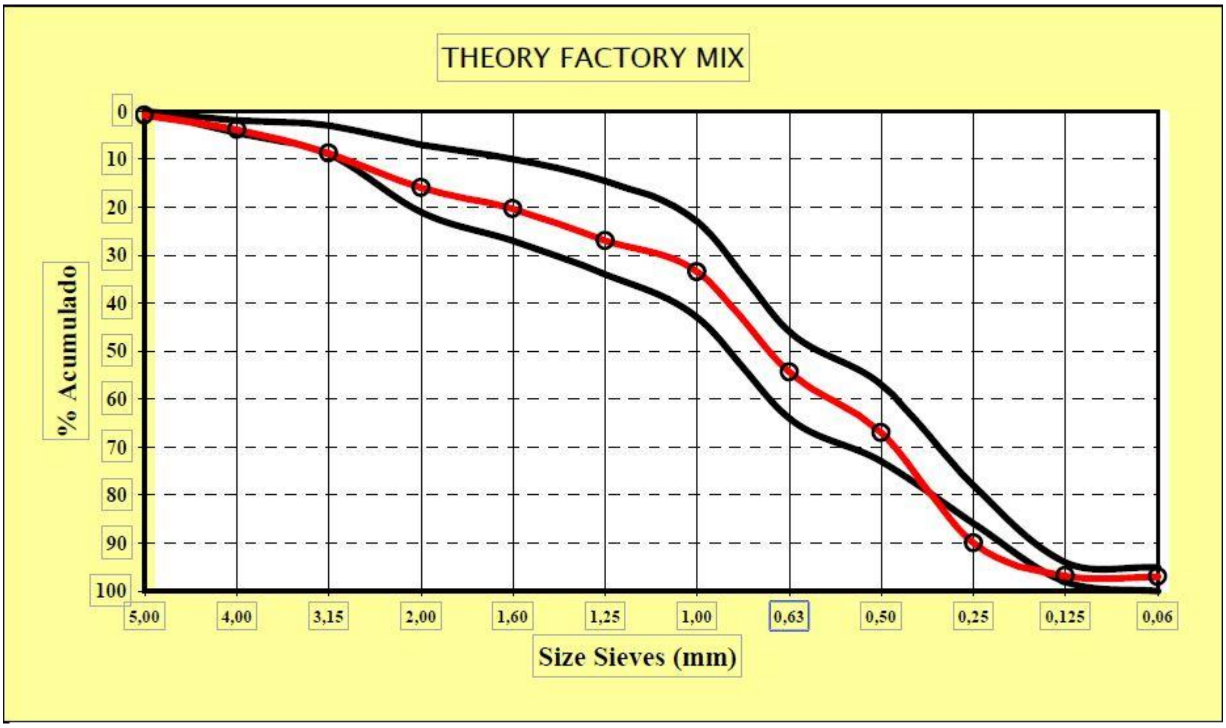
The stone fragments unacceptable : **Calcium, Clay, Soft irestone / Silstone**

THE THEORETICAL IDEAL SAND

In my opinion, the possible theoretical ideal sand for the production should be of the following granulometry.

nº Sieve.		Size in mm.	%	% of the total
	4	4,76	0	0
	8	2,38	10	10
	12	1,7	17	7
	116	1,19	27	10
	30	0,59	57	30
	50	0,297	78	21
	100	0,149	94	16
	200	0,074	97	3
Powder		0,05		3
				100

An proposal of a theoretical mix of 3 existing sands (gross, fine and crushing)



GRANULOMETRY.

PREFIXED OBJECTIVES :

Fineness Modul ………(2,7 to 3%)
Fines 1 …….………….(2 to 5 %)
Sand Equivalent …….(80 to 85 %)

FINENESS MODUL

The calculation is made by adding the retained percentages of the sieves.

4+2+1+0,5+0,25+0,125 and dividing them by 100.

FINES 1

It is the loss of fines that passes through the sieve of 0.0623 mm with the help of water (it is called fine washing)

The percentage of fines lost due to this washing is valued as a percentage

GRANULOMETRY.

SAND EQUIVALENT.

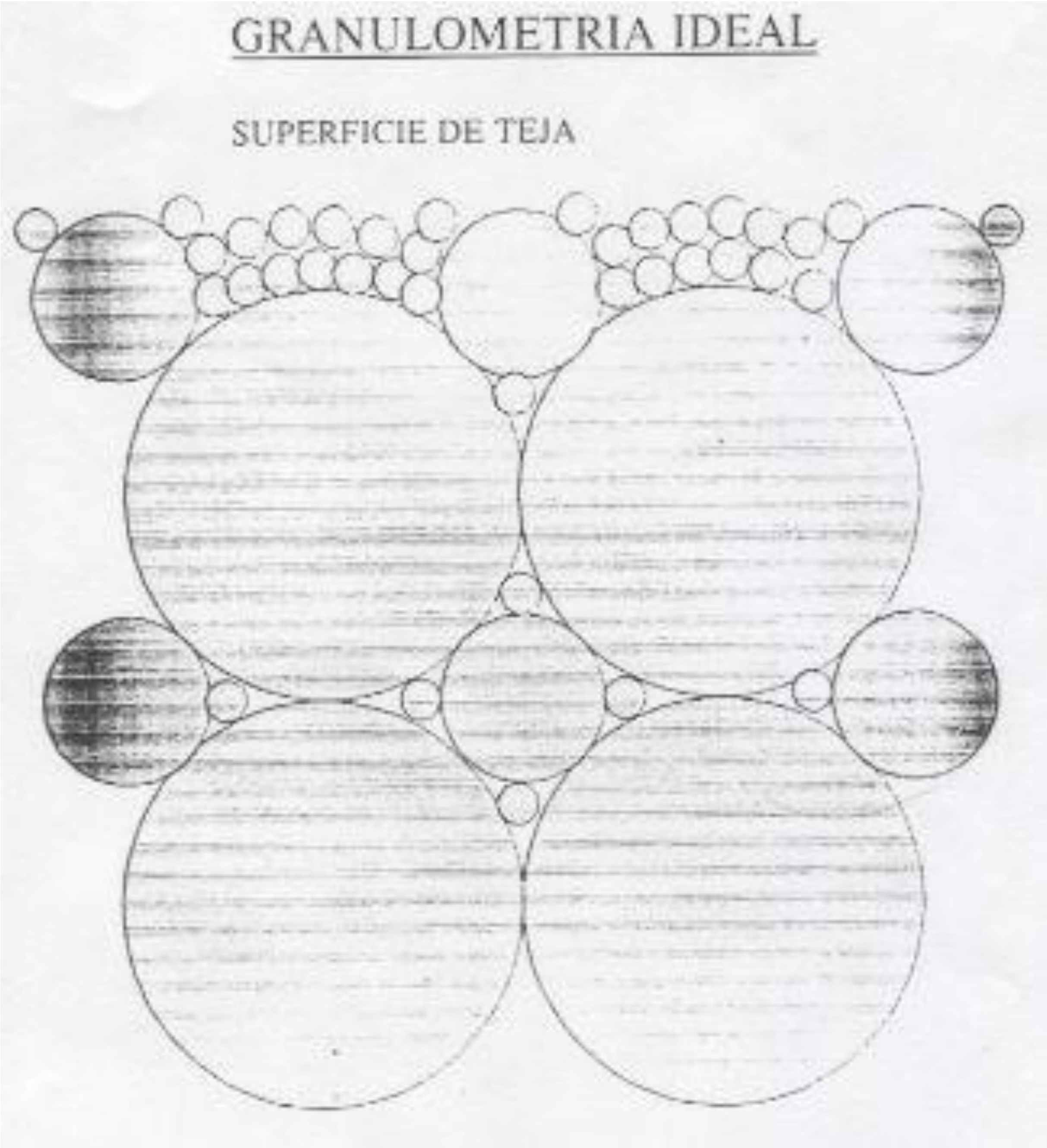
The sand **equivalent test indicates the ratio between the granular and clay elements of an aggregate (sand).**

Sand Equivalent (EA) = (Sand measuring / Clay measuring) x 100

There is very little difference between the values sand and the upper part of the clay.

GRANULOMETRY.

IDEAL SAND



MIX OF SANDS.

As a reference of sand mixtures we can indicate 2 real mixtures in different factories with different proportions and classes of sands.

Sand 0/5 605 kg. + 0/5 160 kg. + 0/2 Crushing Sand 135 kg. = 900 kg.

Máx = 15%

Sand 0/4 503 kg. + 0/4 504 kg. + 0/2 271 kg. = 1.278 kg.

For each profile and model you will have different amounts of sand in these 2 factories

3,443 kg / unit

3, 448 kg / unit

3,450 kg / unit

3,625 kg / unit

There may be dispersions by thickness of the tile or by the mold itself.

CEMENT

As a reference for Cement weights, we can indicate actual cement weighings in different factories with different proportions and possibilities of cement.

The cement weights in the mix for a factory that doubles are usually for the first 2 or 3 hours of production start in the morning.

NORMAL CURING   220 KG. I 42,5R PÓRTLAND.

SHORT CURING   115 KG. I 42,5R PÓRTLAND + 115 KG. I 52,5 PÓRTLAND

NORMAL CURING 350 KG./ 380 KG I 42,5R PÓRTLAND.

SHORT CURING 280 / 310 KG I 42,5R PÓRTLAND.+ 40 / 70 KG.I 52,5 PÓRT.

The cement weights in the tile are reference to a type of mold and model that can be optimized or not.

0,973 kg / unit

0,943 kg / unit

0,942 kg / unit

0,932 kg / unit

Depending on the moulds there is a greater or lesser use of cement, as a guideline as a reference:

	Kg / unit
	0,877
Wave S.	0,913
	1,045

ADDITIVES.

CHEMICAL PRODUCTS

Many years ago, we added cement, fly ash, steel slag, by-products, etc.

The cement companies were added to their cement recipe.

PLASTICIZING

DO NOT use chemical products, powders or liquids in the tile mix that can alter the structural characteristics (to achieve a greater initial resistance or a greater plasticity of the mortar) (accelerators, stiffeners, setting retarders, etc.) so far in my experience I have not identified that provide more advantages, on the contrary if I have identified disadvantages in its use.

(There may be other criteria in this regard, which I respect but do not share).

In factories for surface finishes such as slurry SI, small doses are used for the formation of a 1 mm layer and alter its characteristics.

WATERPROOFING

Add elements / components for a contribution of fines (if the sand used did not have enough fines) whose objective was to increase and solve problems of impermeability without other purpose and not to alter the structure of the resulting concrete.

I consider it necessary to add a little waterproofing to guarantee waterproofing.

I can only contrast the use of calcium stearate and zinc powder has been my experience in this product because we had a supplier and a good price.

In additions to the manual mixer were made by volume.

With values of 0.822 grams / tile to a maximum (exceptional) of 4 grams.

As means 0,894, 1,5, 1,8, 2,7 grams / unit according to factories and profiles.

The one that requires more waterproofing by profiles, is the flat tile.

The waterproofing tests for the guarantee certificate give the evaluation of the waterproofing addition needs,

MIXER

MIXER CYCLES

Check and ensure if the cycles of the mixer for the dry and wet mortar are sufficient for a good cycle in the mixer for the good homogenization of the mortar.

It is necessary to achieve good homogenization without lumps.

There are 2 times, one of discharge of sand, cement, additives and water that depend on the systems of addition and others that depend on the energy that brings the mixer in the dry mix (sand , cement, pigment, additives) that depend on the systems of addition and others that depend on the energy provided by the mixture in the dry mix and the wet mix (sand, cement, pigment, additives + water)

All usually mixers have a humidity / moisture meter,

1) The test time for sand humidity 25 "

2) Dry mix 30 "/ 40"

3) The wet mix 50 "/ 70"

Total 150 "/ 170"

100% Mixer capacity at 105 t.p.m x 4.6 kg x 180 "= 1500 kg of minimum mixture.

The mixing time is given by the good homogenization in the mixer is given by the characteristics of the mixer.

Maximum capacity for 1,400 kgs. 60 tpm.

Minimum Kneading time .... Cycle 195 "(SCHLOSSER)

For a factory at 60 / 85 tiles per minute.

RATIO SAND / CEMENT

The ratio used of sand / cement most used in Spain, depending on whether the tile mould is optimized or not .

3,2 to 3,9

Ratio the reference in Italy ………...……… .3,3

Example recipe Mix (Sand/Cement):

Fine Sand 0/2 ……..360 kgs.

Gross Sand 0/5 ...…740 kgs.

Cement.....…………300 kgs.

With the best mix of sands, cement can be reduced, but it will always indicate it the strength of the tile in the resistance test.

Fine sand look better, thicker sand improves the resistance.

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MIXTURE MOISTURE

The factories were started manually, the operator added the water, after with a cuadalimeter to then automatically add a few programmed liters and a final adjustment manualy.

At present, the moisture of the concrete is monitored by an automatic sensor system in the mixer.

What is the ideal moisture of the mixture in the mixer?

The moisture of the ideal mortar that we look for in our conditions are from 8% to 9%. (the objective is generally 8.5%)

It can reach up to 10% in very special cases of sand and tile profile, it is good identify the moisture in the mixture.

EXAMPLE RECIPE MIXTURE:

Sand / Cement = 3.66

Fine Sand 0/2 ... 360 kgs.

Gross sand 0/5 ... 740 kgs.

Cement .... 300 kgs.

Water .... 60 to 65 liters.

Moisture mortar from 8.3% to 9%.

Mix time………… ....195 " (Depending on the type of mixer)

There is a ratio that is used in concrete for building, it is water / cement.

PIGMENT

Always from the beginning of the factory has worked with powder pigment manually.

*As a reference for pigment weights, we can indicate actual weighings of pigments in different factories with different proportions and possibilities of pigment for each tile color.

In order to match the tile colors of all the factories, we must compensate different color of the sands and cement.

EXAMPLES OF PIGMENT PERCENTAGES AND WEIGHTS IN THE TILE.

P / C: Grey 3,18 %.

P / C: Brown 3,64 %.

P / C: Grey 4,55 % (Flat tile by color intensity).

P / C: Grey 2,8 %.

P / C: Brown 2,7 %.

27,613 g / unit

25,161 g / unit

18,848 g / unit

17,38 g / unit

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